

Bluetooth Channel Sounding

Mode-0 Frequency and Timing Calibration

What is actually observable when two free-running oscillators measure each other, why actuation error must be exchanged, single-tone estimation over $80\ \mu\text{s}$, and what the ranging modes really need from the result

This report is an independent treatment of Channel Sounding Mode-0, written from first principles rather than derived from earlier drafts. It starts from the only thing a receiver physically observes — complex phase rotation against its own reference, timed by its own clock — and builds up to the fractional frequency offset that the ranging modes consume. It separates three quantities that are routinely conflated: the *oscillator* rate offset between the two devices, which is what scales time and phase downstream; the *actuation* error of each synthesiser, which is a resolution artefact and is why frequency actuation error has to be communicated between the devices at all; and the *baseband beat* that is the only thing actually measured. It derives the single-tone estimator family and its Cramér–Rao bound, shows that the thermal floor over the $80\ \mu\text{s}$ tone is about 0.03 ppm, and then shows that this floor is not the binding constraint — a sign convention, a DC offset, or an untrimmed settling transient each cost more than the noise does. It closes by quantifying, against the companion Mode-1, Mode-2 and Mode-3 reports, exactly how much frequency accuracy each downstream mode requires, and therefore when Mode-0 is good enough and when it is not.

Primary references	Core 6.3, Vol. 6, Part A, Sec. 3.5; Vol. 6, Part H, Sec. 4.3.1
Architecture	Core 6.3, Vol. 1, Part A, Secs. 9.1–9.2
Host interface	Core 6.3, Vol. 4, Part E, LE CS Subevent Result event
Companion reports	<i>Mode-1 RTT Estimation</i> , Rev. 1.0; <i>Mode-2 Phase-Based Ranging</i> , Rev. 1.0; <i>Mode-3 Combined RTT and PBR</i> , Rev. 1.0
Relationship to prior work	Independent; supersedes nothing. See §1.
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Abstract

Mode-0 is the calibration step of Bluetooth Low Energy Channel Sounding [Core 6.3, Vol. 6, Part H, Sec. 4.3.1]. The initiator transmits a `CS_SYNC` packet; after the interlude period the reflector transmits its own `CS_SYNC`, a guard interval, and then an unmodulated tone of fixed duration $T_{FM} = 80\ \mu s$. The initiator measures that tone. From it the initiator derives two outputs that every later step in the subevent depends on: a frequency scale, expressed as a fractional frequency offset, and a timing reference that tells it when the reflector's subsequent transmissions will occur.

The physics is simpler than the terminology suggests, and this report tries to keep the two apart. A receiver cannot observe radio frequency. It observes a phase that rotates at the difference between the incoming carrier and its own local oscillator, sampled on its own clock. That difference contains two physically distinct contributions. The first is the *oscillator rate offset* between the two devices — a dimensionless quantity that scales every frequency and every time interval on the link, and the only thing the ranging modes actually need. The second is the *actuation error* of each device's synthesiser: a real synthesiser cannot emit exactly the commanded frequency, so it emits the nearest value it can reach, and the residual is an artefact of frequency resolution with no connection to clock rate. Because the two contributions are inseparable within a single measurement, and because each device knows only its own actuation error, the peer's value must be communicated — which is precisely why frequency actuation error exists as an exchanged quantity in the specification, and why it must be removed rather than averaged away.

The report derives the reconstruction from measured baseband beat to received radio frequency, giving explicit attention to the in-phase/quadrature sign convention, whose inversion silently doubles the error rather than removing it. It develops the estimator family — lag- m autocorrelation, unwrapped-phase least squares, and periodogram refinement — with their ambiguity limits, and shows that a lag-64 estimator at a 4 MHz sample rate wraps at ± 12.8 ppm, which is inside the tolerance of a legitimate crystal pair. A Cramér–Rao analysis gives a thermal floor of 0.032 ppm at 20 dB over the full tone, degrading to 0.043 ppm once realistic settling and tail margins are trimmed. An error budget then shows that this floor is comfortably below what any downstream mode requires, and that Mode-0 accuracy in practice is decided by systematic effects rather than by noise.

Status and interpretation. Normative claims are tied to the cited Bluetooth Core Specification sections and have been checked against the adopted Core 6.3 text: the fractional frequency offset relation, the actuation-error semantics and encoding, the step structure and every timing parameter used here are quoted or derived from it. Estimators, windowing, the in-phase/quadrature sign convention, quality gates and calibration schemes are not specified and are representative implementation methods. Three numerical inconsistencies in the surrounding series material, all now settled against the specification, are recorded in Section 3. The Bluetooth specification and adopted test specifications remain authoritative for conformance.

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1 Scope, and the relationship to the earlier Mode-0 report

1.1 Relationship to prior work

A previous Mode-0 report exists in this series at Revision 4.0. This document was written independently and from first principles, at the author's request, and does not modify, correct or supersede it. The two are complementary rather than competing: where the earlier report builds a formal reference-frame apparatus and carries a fully instrumented numerical example through every intermediate value, this one aims to make the physical content legible first and to state plainly which parts are specified, which are derived, and which are uncertain.

The two agree on the central reconstruction. Working independently, this report arrives at Eq. (6), which is the same relation the earlier report states in its abstract. That agreement is reported here because it is evidence, not because it is a citation: two derivations from different starting points reaching the same equation is the strongest check available in the absence of the specification text itself.

1.2 What Mode-0 is for

Mode-0 is not a distance measurement and produces no ranging observable. It exists so that the steps after it in the same subevent can assume a known frequency and timing relationship between the two devices [Core 6.3, Vol. 1, Part A, Sec. 9.1]. It has two outputs, and reports that name themselves after only the first tend to under-serve the second.

1. **A frequency scale.** The fractional frequency offset between the two devices' references. Mode-1 uses it to rescale the reflector's reported turnaround interval; Mode-2 and Mode-3 use it to bound the phase that drifts between their reciprocal measurements; all three use it to place the expected carrier so that the peer's transmission lands where the receiver is listening.
2. **A timing reference.** The reflector's CS_SYNC in the Mode-0 step tells the initiator when the reflector's transmissions actually occur, against the initiator's own clock. Every subsequent step in the subevent is scheduled against that reference, and the receive windows of later steps are opened on the strength of it.

Why the calibration is repeated per subevent. Neither output is a constant. Reference oscillators drift with temperature and supply, and the relationship between two of them drifts faster than either alone. A subevent begins with one to three Mode-0 steps rather than a procedure-level calibration because the correction must be fresh: what matters is the offset during *these* ranging steps, not the offset when the procedure started.

1.3 Questions this report answers

1. What, exactly, does a receiver physically observe during the Mode-0 tone?
2. Which part of that observation is the quantity the ranging modes need, and which part is an artefact that must be removed?
3. Why does frequency actuation error have to be exchanged between the devices, rather than each device handling its own?
4. How does phase rotation per sample become a fractional frequency offset, and where does the sign come from?
5. How precisely can the offset be estimated over $80\ \mu\text{s}$, and what limits it in practice?

6. How much accuracy does each downstream mode actually require, and does Mode-0 deliver it?

2 Normative framework and terminology

Table 1: Normative map for Mode-0. Section numbering should be confirmed against the specification revision in force.

Source	What it defines	Question answered
Core 6.3, Vol. 1, Part A, Sec. 9.1	CS procedure architecture; the calibration role of the first steps	Why does a subevent begin with Mode-0?
Core 6.3, Vol. 6, Part H, Sec. 4.3.1	Mode-0 step structure and ordering	What is transmitted, by whom, in what order?
Core 6.3, Vol. 6, Part H, Sec. 4.3	T_{SY} , T_{GD} , T_{FM} , T_{RD} , T_{IP1} and permitted values	Which step timings apply?
Core 6.3, Vol. 6, Part A, Sec. 3.5	Frequency measurement and generation in CS; $f_{RX}[k]$, actuation error, the FFO relation	What is the normative definition of the measured quantity?
Core 6.3, Vol. 6, Part H, Sec. 4.4	CS_SYNC packet construction	What waveform provides the timing reference?
Core 6.3, Vol. 6, Part B	CS link-layer control procedures, including the exchange of actuation-error data	How does the peer's actuation error reach the receiver?
Core 6.3, Vol. 4, Part E, LE CS Subevent Result event	Per-step Mode-0 reporting	How is the result exposed to the host?
RFPHY.TS / LL.TS, CS Mode-0 clauses	Instrument-level verification	How is the behaviour tested?

Table 2: Principal symbols. The decomposition in the middle block is this report’s; the specification’s own naming should be preferred where they differ.

Symbol	Meaning	Units
k	Mode-0 step index within the subevent	—
$f_{nom}[k]$	Nominal (commanded) centre frequency of the step’s channel	Hz
$\varepsilon_I, \varepsilon_R$	Fractional rate error of each device’s reference oscillator	—
ε	Relative oscillator offset $\varepsilon_R - \varepsilon_I$; the quantity sought	—
δ_I, δ_R	Frequency actuation error of each synthesiser, in its own frame	Hz
$f_{LO}^{(I)}[k]$	Initiator’s actual local-oscillator frequency	Hz
$f_{TX}^{(R)}[k]$	Reflector’s actual transmitted tone frequency	Hz
$\Delta f_{BB}[k]$	Measured baseband beat frequency	Hz
s_{IQ}	In-phase/quadrature sign convention, ± 1	—
$f_{RX}[k]$	Receiver’s estimate of the average received tone frequency	Hz
F AE	Frequency actuation error as communicated between devices	Hz or ppm
FFO[k]	Fractional frequency offset from step k	ppm
FFO*	Subevent fractional frequency offset after combining steps	ppm
$z[n]$	Received complex baseband sample	ADC units
F_s	Receiver sample rate (calibrated)	Hz
T_{obs}	Accepted observation window inside T_{FM}	s
σ_f	Standard deviation of the frequency estimate	Hz

3 Numerical inconsistencies in the existing series material

Three parameter conflicts in the surrounding series material were identified while writing this report and have since been settled directly against Core 6.3. All three are resolved in favour of the values used here; in every case the timing artwork held with the series is the outlier and should be corrected.

(1) T_{SY} for a Mode-0 packet: $16\mu s$ or $26\mu s$? The Mode-0 timing artwork held with this series states $T_{SY} = 16\mu s$ in its legend. That value is not reachable from the CS_SYNC packet construction used in the Mode-1 report, which gives, for a packet carrying no sounding or random sequence, $8 + 32 + 4 = 44$ bits on the LE 1M PHY ($44\mu s$) and $16 + 32 + 4 = 52$ bits on the LE 2M PHY ($26\mu s$). Sixteen microseconds corresponds to 32 bits at 2 Mb/s — the access address alone, with neither preamble nor trailer.

Resolved against the specification. Core 6.3 states that “CS packets with no Sounding Sequence or Random Sequence fields take $44\mu s$ to transmit when sent using the LE 1M PHY and $26\mu s$ to transmit when using the LE 2M and the LE 2M 2BT PHYs” [Core 6.3, Vol. 6, Part H, Sec. 2], and separately that “the CS_SYNCS for mode-0 shall not carry a sounding sequence or random sequence” [Core 6.3, Vol. 6, Part H, Sec. 4.3.1]. T_{SY} for a Mode-0 packet is therefore $26\mu s$ on LE 2M and $44\mu s$ on LE 1M. The artwork’s $16\mu s$ is wrong.

(2) T_{GD} : $3\mu s$ or $10\mu s$? The same artwork legend gives a guard delay of $3\mu s$, while the earlier Mode-0 report and the campaign report both use $10\mu s$.

Resolved against the specification. Core 6.3 states that “the guard time (T.GD) is $10\mu s$ in duration, independent of the LE PHY being used” [Core 6.3, Vol. 6, Part H, Sec. 2.6]. The artwork’s $3\mu s$ is wrong. This also confirms the companion Mode-3 report’s correction to the campaign report’s step duration, which is $2T_{GD} = 20\mu s$.

(3) **The name of the interlude.** The artwork expands $T_{IP,1}$ as “instantaneous phase interval”. The interval is an idle period between the two devices’ transmissions and carries no phase measurement; the Mode-1 report calls it the *interlude period*, which matches its function. Core 6.3 describes $T_{IP,1}$ as “the idle time between the end of the transmission from the initiator and the start of the transmission from the reflector” [Core 6.3, Vol. 6, Part H, Sec. 4.3.1] — an idle period, with no phase observable in it. This report uses “interlude period”. The distinction is terminological, not numerical, but a reader who takes the artwork’s expansion literally will look for a measurement that does not exist.

Table 3: Mode-0 step duration under the competing parameter claims, from Eq. (1) with $T_{RD} = 5\mu s$, $T_{FM} = 80\mu s$ and $T_{IP,1} = 40\mu s$. The first row is the specification-confirmed combination.

Parameter source	T_{SY}	T_{GD}	$T_{step,0}$
Core 6.3, LE 2M PHY [Core 6.3, Vol. 6, Part H, Secs. 2, 2.6]	$26\mu s$	$10\mu s$	$192\mu s$
Timing artwork legend (incorrect)	$16\mu s$	$3\mu s$	$165\mu s$
Core 6.3, LE 1M PHY	$44\mu s$	$10\mu s$	$228\mu s$

4 Step structure and timing

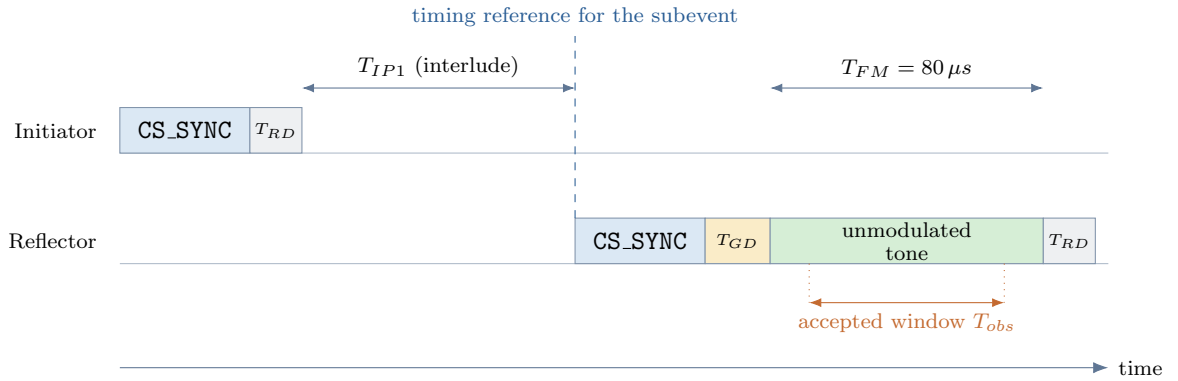


Figure 1: Mode-0 step. The initiator’s CS_SYNC lets the reflector align; the reflector’s CS_SYNC gives the initiator its timing reference for the whole subevent, and the $80\mu s$ unmodulated tone that follows gives it the frequency scale. Only the initiator makes a frequency measurement. The estimator works on a window strictly inside T_{FM} , trimmed at both ends (Section 6).

The nominal step duration is

$$T_{step,0} = 2T_{SY} + 2T_{RD} + T_{IP,1} + T_{GD} + T_{FM}, \quad (1)$$

which is the expression given in the specification itself [Core 6.3, Vol. 6, Part H, Sec. 4.3.1], with $T_{FM} = 80 \mu s$, $T_{GD} = 10 \mu s$ and $T_{RD} = 5 \mu s$ fixed and T_{IP1} configured from the set $\{10, 20, 30, 40, 50, 60, 80, 145\} \mu s$, of which $145 \mu s$ is mandatory [Core 6.3, Vol. 6, Part H, Table 4.5]. A Mode-0 packet carries no sounding or random sequence, so T_{SY} takes only its shortest value: $26 \mu s$ on the LE 2M PHY and $44 \mu s$ on LE 1M.

The step is deliberately asymmetric. Core 6.3 states that “the CS initiator performs a measurement of the frequency offset of the transmitted RF signal from the CS reflector during CS step mode-0”, and that it “shall use this information to compensate for the frequency and timing error contributions in all further transmissions within subsequent CS steps in a CS subevent” [Core 6.3, Vol. 6, Part H, Sec. 4.3.1] — which also confirms the two outputs of Section 1. Unlike every other CS step mode, Mode-0 is not reciprocal. The reflector transmits a tone and the initiator measures it; the reverse measurement is never made. This is sufficient because the quantity of interest is a *relative* offset, which is antisymmetric in the two devices: measuring $\varepsilon_R - \varepsilon_I$ once determines $\varepsilon_I - \varepsilon_R$ as well. It also means the whole of Mode-0’s estimation burden falls on the initiator, and that a reflector-side receiver defect will not show up in Mode-0 at all — it will show up later, as a Mode-1 or Mode-2 error with no obvious cause.

4.1 Number of steps per subevent

A subevent begins with one to three Mode-0 steps. More steps buy two different things, and it is worth separating them: they reduce the variance of the frequency estimate as $1/\sqrt{M}$, and they give the receiver more than one chance to acquire the timing reference under interference. A design that is timing-limited and a design that is noise-limited will choose the count for different reasons.

5 What is actually observable

This section builds the measurement from the physics up. It is the core of the report, and everything later is either estimation theory applied to Eq. (5) or engineering applied to Eq. (7).

5.1 Two oscillators

Each device derives everything it does — its carrier frequencies and its clock ticks alike — from one reference oscillator. Let device X ’s reference run at a fractional rate error ε_X relative to an ideal reference. This single number scales two things at once, and that coupling matters later: a device whose reference is fast by ε_X transmits at frequencies high by the factor $(1 + \varepsilon_X)$ and measures time intervals short by the same factor.

Define the relative offset

$$\varepsilon \equiv \varepsilon_R - \varepsilon_I. \tag{2}$$

Only ε is observable, and only ε is needed. Neither device can determine its own ε_X : doing so would require an absolute frequency standard, which neither has. Nor does either need to. Every downstream use is a comparison between the two devices — rescaling the reflector’s reported interval onto the initiator’s clock, predicting how far the relative oscillator phase drifts between two measurements, placing a carrier where the peer expects it — and every one of those depends on the difference alone. Mode-0 is not a frequency-accuracy measurement; it is a frequency-*comparison* measurement, and that is why one asymmetric exchange suffices.

5.2 Actuation error: why the commanded frequency is not the emitted one

A synthesiser cannot emit an arbitrary frequency. It is built from a reference and a divider structure with finite resolution, so when it is commanded to produce $f_{nom}[k]$ it produces the nearest value its architecture can reach. Write that residual, expressed in the device's own reference frame, as the actuation error $\delta_X[k]$. The actual frequencies are then

$$f_{TX}^{(R)}[k] = (1 + \varepsilon_R)(f_{nom}[k] + \delta_R[k]), \quad (3)$$

$$f_{LO}^{(I)}[k] = (1 + \varepsilon_I)(f_{nom}[k] + \delta_I[k]). \quad (4)$$

Two properties of δ_X drive the whole design. It is *channel-dependent*, because which frequencies are reachable changes across the band, so it is a table over channels rather than a constant. And it is *locally knowable*: a device commanded to a frequency knows what its own synthesiser actually produced, to far better accuracy than it could measure the peer.

5.3 The beat, and what is in it

The initiator downconverts the incoming tone against its own local oscillator and observes a complex baseband signal whose phase rotates at the difference frequency. Subtracting Eq. (4) from Eq. (3),

$$\begin{aligned} \Delta f_{true}[k] &= (1 + \varepsilon_R)(f_{nom} + \delta_R) - (1 + \varepsilon_I)(f_{nom} + \delta_I) \\ &= f_{nom} \varepsilon + (\delta_R - \delta_I) + \underbrace{(\varepsilon_R \delta_R - \varepsilon_I \delta_I)}_{\text{second order}}. \end{aligned} \quad (5)$$

The second-order term is the product of a part-per-million quantity and a synthesiser residual of at most a few hundred hertz; it is below a millihertz and is dropped throughout.

The one equation that explains the whole of Mode-0's terminology. Equation (5) says the measured beat is the sum of two physically unrelated things. The first, $f_{nom}\varepsilon$, is the oscillator rate offset scaled by the carrier — the quantity every ranging mode needs. The second, $\delta_R - \delta_I$, is a difference of synthesiser resolution artefacts that has nothing to do with clock rate and must not be allowed to masquerade as one. A single measurement cannot separate them, because they enter identically. Therefore they must be separated by *knowledge*, not by estimation — and since the initiator knows only δ_I , the reflector must tell it δ_R . That exchange is what frequency actuation error is for.

5.4 Reconstruction: from baseband beat to received frequency

What the receiver measures is a baseband rotation rate, $\widehat{\Delta f}_{BB}[k]$, in hertz on its own time-base. The specification's measured quantity, however, is $f_{RX}[k]$: the receiver's estimate of the average *received* frequency of the tone [Core 6.3, Vol. 6, Part A, Sec. 3.5]. Recovering one from the other requires adding back the local oscillator the receiver mixed against, and getting the sign right:

$$\widehat{f}_{RX}[k] = f_{nom}[k] + \delta_I[k] + s_{IQ} \widehat{\Delta f}_{BB}[k] \quad (6)$$

where $s_{IQ} = \pm 1$ encodes whether the receiver's quadrature convention produces an erect or inverted baseband spectrum. Note what appears in Eq. (6): the receiver adds its own *actual* local-oscillator frequency, which is the commanded value plus its own actuation error. It does not add the nominal channel frequency.

The sign convention is the single most damaging error in Mode-0. An inverted s_{IQ} does not attenuate the correction; it reverses it. A +12 ppm offset is then reported as −12 ppm, and every downstream correction is applied backwards, doubling the error it was meant to remove. The failure is silent: the estimator’s variance, its residuals, and its quality metrics are all unchanged, because the magnitude is right. It is detectable only by an end-to-end test with a deliberately injected offset of *known sign* — which is why that test appears in every verification plan in this series, and why it should be run before any accuracy characterisation, not after.

5.5 From received frequency to fractional frequency offset

The fractional frequency offset is the beat expressed dimensionlessly, with both actuation errors removed. Combining Eqs. (5) and (6),

$$\widehat{\text{FFO}}_{RX}[k] = 10^6 \frac{\widehat{f}_{RX}[k] - f_{nom}[k](1 + 10^{-6}\text{FAE}[k])}{f_{nom}[k](1 + 10^{-6}\text{FAE}[k])} \text{ ppm} \approx \varepsilon, \quad (7)$$

which is the relation given in Core 6.3, Volume 6, Part A, Section 3.5.1 [Core 6.3, Vol. 6, Part A, Sec. 3.5.1]. Here $\text{FAE}[k]$ is the peer’s actuation error in parts per million, so that the δ_R of Eq. (5) is recovered as

$$\delta_R[k] = 10^{-6} \text{FAE}[k] f_{nom}[k]. \quad (8)$$

Substituting Eq. (6) into Eq. (7) and expanding to first order gives the equivalent and more transparent form

$$\widehat{\text{FFO}}_{RX}[k] \approx \frac{s_{IQ} \widehat{\Delta f}_{BB}[k] - (\delta_R[k] - \delta_I[k])}{f_{nom}[k]}, \quad (9)$$

which makes plain that the estimate is the measured beat with the actuation-error difference subtracted before normalisation.

Normalise by the adjusted frequency, not the nominal one. The denominator in Eq. (7) is $f_{nom}(1 + 10^{-6}\text{FAE})$, not f_{nom} . The difference is a relative factor of at most 4×10^{-6} , so on an offset of 50 ppm — the specified limit — it changes the result by about 2×10^{-4} ppm. That is negligible against every other term in this report, but Eq. (7) is exact and Eq. (9) is not; an implementation should compute the former.

Verified against Core 6.3. Equation (7) reproduces the specification’s $\text{FFO}_{RX}[k]$ relation exactly [Core 6.3, Vol. 6, Part A, Sec. 3.5.1]. Three further points of this section are confirmed by the same text. The specification states that “the FAE is communicated by the reflector”, which is the normative basis for Section 5’s central argument that the peer’s actuation error must be exchanged rather than estimated. It defines $f_{RX}[k]$ as “the receiving device’s estimate of the average frequency of the received CS tone”, and requires that “the receiving device should compensate for its own local frequency actuation error (LFAE)” as part of that estimate — which is precisely Eq. (6). And Section 6.3 defines the LFAE as “the frequency offset between the generated frequency and the expected frequency”, which is the δ_I of Eq. (4). The in-phase/quadrature sign convention s_{IQ} is not a specification matter and remains an implementation concern.

5.6 The exchanged actuation error is quantised

The peer’s actuation error does not arrive as a real number. It is carried in the **ChFAE** field of the **LL_CS_FAE_RSP** PDU as a per-channel table of 8-bit signed integers [Core 6.3, Vol. 6, Part B, Sec. 2.4.2.52]. Each entry spans $[-4, +3.96875]$ ppm with a resolution of 0.03125 ppm: the value is the actuation error multiplied by 32 and rounded, so −128 represents −4 ppm and +127 represents +3.96875 ppm. Only the allowed CS channels are represented, shifted down to fill the indices left void by channels that are not permitted.

Two consequences follow, and the second is easy to miss.

Quantisation of FAE is a bias, not noise. The rounding error on a channel's table entry is fixed for that channel: it is the same in every step, every subevent and every procedure on that link. It therefore does *not* average down, and no increase in the Mode-0 step count reduces it. The worst case is half a least-significant bit, 0.015625 ppm — which is 38 Hz at 2.44 GHz, about half the thermal floor of Section 7.4, and 28% of the binding downstream requirement derived in Section 10. Treated as a uniform random variable across channels its root-mean-square value is $0.03125/\sqrt{12} = 0.0090$ ppm.

The full-scale range matters as much as the resolution. Because FAE may legitimately reach 4 ppm, an implementation that fails to apply the table at all does not incur the small error of the worked example in Section 11; it incurs up to 4 ppm, which is more than seventy times the binding downstream requirement and some 11 cm of Mode-2 range.

5.7 Why the sample clock does not need separate correction

A tempting worry is that the receiver measures Δf_{BB} against its own sample clock, which is itself in error by ε_I , so the measurement must be rescaled. The worry is correct in principle and negligible in magnitude, and it is worth settling numerically so that nobody over-engineers it.

The measured beat is the true beat divided by the local timebase scaling, so the fractional error introduced is ε_I applied to a quantity that is itself small:

$$\text{error} = f_{nom} \varepsilon \varepsilon_I. \quad (10)$$

At $f_{nom} = 2.44$ GHz with both ε and ε_I at a generous 12 ppm, this is 0.35 Hz — against a thermal floor of some 77 Hz (Section 7.4). It is two orders of magnitude below the noise and can be ignored.

But the sample rate itself must still be calibrated. Equation (10) concerns the tiny second-order coupling through ε_I . It says nothing about a receiver whose nominal sample rate is simply wrong — a decimation factor mis-set, a nominal F_s used where the physical rate differs. That is a first-order scale error on $\widehat{\Delta f_{BB}}$ and it maps directly into FFO. The estimator must convert cycles-per-sample to hertz using the calibrated physical rate, not the nominal one.

6 From radio frequency to complex baseband

Section 5 worked in frequencies: the reflector emits at $f_{TX}^{(R)}$, the initiator's oscillator sits at $f_{LO}^{(I)}$, and the difference is the beat. This section carries that through the receiver, because two quantities the earlier sections simply asserted — the complex baseband model and the sign s_{IQ} — are properties of the mixing chain and are best derived rather than assumed.

6.1 Quadrature downconversion

During T_{FM} the reflector emits an unmodulated carrier, which arrives at the initiator's antenna as the real passband signal

$$r(t) = A_R \cos(2\pi f_{TX} t + \varphi_{TX}) + n(t), \quad (11)$$

where $f_{TX} \equiv f_{TX}^{(R)}[k]$ of Eq. (3), φ_{TX} is an arbitrary carrier phase and $n(t)$ is the passband noise.

The receiver generates two local-oscillator phases in quadrature at $f_{LO} \equiv f_{LO}^{(I)}[k]$ of Eq. (4) and mixes the incoming signal with each:

$$p_I(t) = 2 \cos(2\pi f_{LO}t + \varphi_{LO}), \quad p_Q(t) = -2 \sin(2\pi f_{LO}t + \varphi_{LO}). \quad (12)$$

Writing $a = 2\pi f_{TX}t + \varphi_{TX}$ and $b = 2\pi f_{LO}t + \varphi_{LO}$ and applying the product-to-sum identities,

$$r(t)p_I(t) = A_R [\cos(a - b) + \cos(a + b)], \quad (13)$$

$$r(t)p_Q(t) = A_R [\sin(a - b) - \sin(a + b)]. \quad (14)$$

Each product contains a difference term near zero frequency and a sum term near $2f_{LO}$. The low-pass filter that follows each mixer removes the sum terms, leaving

$$I(t) = A_R \cos(2\pi \Delta f_{BB}t + \theta_0), \quad Q(t) = A_R \sin(2\pi \Delta f_{BB}t + \theta_0), \quad (15)$$

with

$$\Delta f_{BB} = f_{TX} - f_{LO}, \quad \theta_0 = \varphi_{TX} - \varphi_{LO}. \quad (16)$$

Forming the complex baseband signal in the conventional way,

$$z(t) = I(t) + jQ(t) = A_R e^{j(2\pi \Delta f_{BB}t + \theta_0)}. \quad (17)$$

This is the whole of the measurement. Equation (16) is the link between the frequency algebra of Section 5 and anything a receiver can actually observe: the baseband rotation rate is the difference between the reflector's emitted frequency and the initiator's own oscillator, which by Eq. (5) is $f_{nom}\varepsilon + (\delta_R - \delta_I)$. Note also what Eq. (17) does *not* contain: φ_{TX} and φ_{LO} appear only in the constant θ_0 , never in the rotation rate. Mode-0 needs no phase reference of any kind — which is exactly why it can be asymmetric and why it is unaffected by static multipath.

6.2 Where s_{IQ} comes from

Equation (17) holds for the particular convention of Eq. (12). Real receivers differ, and the differences fall into two genuinely independent categories. Listing them as one flat set invites the confusion this subsection exists to prevent, because the first category is a $\pm 90^\circ$ convention and the second is a property of the analog architecture.

(a) The quadrature convention: a $\pm 90^\circ$ choice

In Eq. (12) the quadrature reference is $-2 \sin(\cdot) = 2 \cos(\cdot + 90^\circ)$, so the Q reference *leads* the I reference by 90° , and Eq. (17) rotates positively. Choosing the other 90° relationship conjugates the baseband signal. Three implementation details express that one choice, and their algebra is not identical:

The first two rows are not two mechanisms but one, reached in hardware and in software respectively: both produce $z = z_{id}^*$ exactly, with no constant term. The channel exchange is different. Substituting $I = \cos \phi$, $Q = \sin \phi$ into $z = Q + jI$,

$$z = \sin \phi + j \cos \phi = j(\cos \phi - j \sin \phi) = j e^{-j\phi} = j z_{id}^*, \quad (18)$$

which is the same conjugation multiplied by a constant $+90^\circ$ rotation.

The constant is harmless here and harmful next door. A frequency estimator measures the *rate* at which z rotates, so the factor j in Eq. (18) is absorbed into θ_0 and Mode-0 cannot tell a channel exchange from an LO polarity inversion — both simply report $-\Delta f_{BB}$. The phase-based modes can: there the reported phase correction term *is* the observable, and a channel exchange puts a fixed 90° into every one of them while an LO polarity inversion puts in none. A defect that is one bug in Mode-0 is therefore two distinct bugs across the series, and a test campaign that verifies s_{IQ} only through Mode-0 will not separate them. Verify the constant as well as the sign.

Table 4: The quadrature convention, and the three places it is decided. Writing $\phi = 2\pi\Delta f_{BB}t + \theta_0$ and $z_{id} = e^{j\phi}$, all three negate the recovered rotation rate — but the channel exchange does so with an additional constant rotation.

Where it is decided	Resulting z	Recovered rate	Constant phase added
Reference (Eq. (12))	z_{id}	$+\Delta f_{BB}$	—
Quadrature LO polarity: +2 sin instead of $-2 \sin$, so $Q \rightarrow -Q$	z_{id}^*	$-\Delta f_{BB}$	none
Firmware assembles $I - jQ$ instead of $I + jQ$	z_{id}^*	$-\Delta f_{BB}$	none
In-phase and quadrature channels exchanged	$j z_{id}^*$	$-\Delta f_{BB}$	$+90^\circ$

(b) The spectral sense of the analog chain

The second category is not a convention but an architectural fact, and it applies only to architectures that pass the signal through a *real* intermediate frequency before the quadrature stage. Each such real mixing stage inverts the spectrum if its local oscillator is above the incoming signal and leaves it erect if below. The net sense is the product over those stages. The direct-conversion receiver derived in Section 6.1 has no real stages at all and contributes +1.

The net convention

Writing $s_{quad} = \pm 1$ for the choice in (a) and $s_m = \pm 1$ for the sense of real mixing stage m ,

$$z(t) = A_R e^{j(s_{IQ} 2\pi\Delta f_{BB} t + \theta'_0)}, \quad s_{IQ} = s_{quad} \prod_{m \text{ real stages}} s_m \in \{+1, -1\}, \quad (19)$$

with θ'_0 absorbing θ_0 and any constant rotation from Eq. (18). The estimator, which measures the rotation rate of z , recovers $s_{IQ}\Delta f_{BB}$ rather than Δf_{BB} ; undoing that is precisely the s_{IQ} factor of Eq. (6).

What is *not* a sign inversion. Whether the *quadrature* mixer’s local oscillator sits above or below the incoming carrier — $f_{LO} > f_{TX}$, so that $\Delta f_{BB} < 0$ — does not belong in either category. A quadrature receiver represents negative baseband frequencies faithfully, and Eq. (17) is already correct with a negative Δf_{BB} ; high-side injection of a quadrature stage changes the sign of the quantity being measured, not the sign of the measurement. Spectral inversion belongs to category (b) and requires a real intermediate frequency. Nor is a decimation or channel-filter chain an inverter.

The consequence is that s_{IQ} cannot be read off a block diagram’s mixer frequencies. It depends on quadrature generation, sample assembly and channel ordering, and an even number of inversions among them cancels invisibly. It has to be *measured*, by the signed injection test of Section 12.

6.3 Sampling, and why Nyquist is not the binding limit

Sampling Eq. (19) at the calibrated rate F_s gives

$$z[n] = A e^{j(2\pi s_{IQ}\Delta f_{BB} n/F_s + \theta_0)}, \quad n = 0, \dots, N-1, \quad (20)$$

which is alias-free provided $|\Delta f_{BB}| < F_s/2$. That condition is comfortably satisfied. The specification bounds the fractional frequency offset on a Mode-0 step at $|\text{FFO}[k]| < 50 \text{ ppm}$ [Core 6.3, Vol. 6, Part A, Sec. 3.5.1], which at 2.44 GHz is 122 kHz, and the actuation-error

difference adds at most a further ± 4 ppm, or 9.8 kHz. The largest legitimate beat is therefore about 132 kHz, some fifteen times inside the ± 2 MHz Nyquist limit of a 4 Msample/s receiver.

The binding ambiguity in Mode-0 is not Nyquist but the lag of the estimator, $F_s/2m$, treated in Section 7 — which for a lag of 64 falls to ± 12.8 ppm, *inside* the legitimate range that Eq. (20) samples perfectly well.

6.4 The received samples with front-end impairments

Two impairments of the mixing chain of Section 6.1 survive into the model and matter enough to be written down. The first is a complex offset d at exactly zero baseband frequency, arising from local-oscillator self-mixing, mixer offsets and converter offsets. The second follows from the two paths in Eq. (12) not being exactly equal: with a gain ratio g and a quadrature phase error ψ between them, the recovered signal is a linear combination of the wanted signal and its own conjugate,

$$z[n] = \alpha z_{id}[n] + \beta z_{id}^*[n] + d + w[n], \quad \alpha = \frac{1 + ge^{-j\psi}}{2}, \quad \beta = \frac{1 - ge^{+j\psi}}{2}, \quad (21)$$

with z_{id} the ideal signal of Eq. (20) and $w[n]$ complex noise of variance σ_w^2 . The conjugate term is the *image*: it sits at $-\Delta f_{BB}$, and its suppression relative to the wanted signal is the image rejection ratio $\text{IRR} = |\alpha/\beta|^2$.

The estimation problem is to recover Δf_{BB} alone. Everything else in Eq. (21) — A , θ_0 , d , β — is a nuisance, and the next subsection shows that two of them are disposed of by one design choice.

Mode-0 is almost immune to multipath, and this is worth knowing. Over a static channel the superposition of reflected copies of a pure tone is another pure tone at the same frequency, with a different amplitude and phase. Multipath therefore changes A and θ_0 in Eq. (21) and leaves Δf_{BB} untouched — a fact that follows directly from Eq. (17), where the carrier phases entered only the constant. This is a sharp contrast with Mode-1, where multipath is the dominant error, and with Mode-2, where it biases the phase slope. Only a *time-varying* channel produces a frequency error, and Section 8 shows that this becomes significant only at vehicular speeds. A Mode-0 implementation should not import multipath machinery from the ranging modes; it has a different problem.

6.5 Intermediate-frequency placement, and why it is the most valuable design choice

The receiver is free to place its local oscillator away from the expected received frequency by a deliberate intermediate frequency f_{IF} , so that

$$\Delta f_{BB} = f_{IF} + f_{nom}\varepsilon + (\delta_R - \delta_I), \quad (22)$$

and the tone no longer sits near zero. This single choice attacks both impairments of Eq. (21) at once: the offset d moves away from the wanted tone, and the image at $-\Delta f_{BB}$ separates from the signal at $+\Delta f_{BB}$ by $2\Delta f_{BB}$.

Its value is easy to underestimate, so Table 5 quantifies it. Both effects are evaluated as the worst case over the unknown phase of the offending term, on an unwrapped-phase least-squares fit over a $65 \mu s$ window.

Table 5: Frequency-estimation error induced by a -30 dBc receiver offset and by a 30 dB image rejection ratio, as a function of where the tone is placed in baseband. Worst case over the phase of the offending term; $T_{obs} = 65 \mu s$; expressed as fractional frequency offset at 2.44 GHz. The thermal floor at the same window and 20 dB is 0.0431 ppm and the binding downstream requirement of Section 10 is 0.055 ppm.

$ \Delta f_{BB} $	DC offset	Image	Combined	Verdict
5 kHz	0.0596 ppm	0.0829 ppm	0.1021 ppm	fails the requirement outright
30 kHz	0.0316 ppm	0.0152 ppm	0.0350 ppm	comparable to the thermal floor
100 kHz	0.0005 ppm	0.0047 ppm	0.0047 ppm	an order of magnitude below it
250 kHz	0.0026 ppm	0.0000 ppm	0.0026 ppm	negligible
500 kHz	0.0000 ppm	0.0009 ppm	0.0009 ppm	negligible

Place the tone at a hundred kilohertz or more, and two error sources disappear. At a nearly zero beat — which is what a receiver gets if it tunes to the nominal channel centre and the two oscillators happen to agree — the combined front-end contribution is 0.1021 ppm, nearly twice the entire Mode-2 requirement and two and a half times the thermal floor. Moving the tone to 100 kHz reduces it to 0.0047 ppm. No estimator refinement, longer window or extra step count buys anything comparable, and the choice costs nothing but a deliberate offset in the local oscillator.

The behaviour is not monotonic, and the reason is worth stating because it explains the worst row. The image contributes a phase perturbation at $2\Delta f_{BB}$; when barely half a cycle of that perturbation fits inside the observation window it looks like a linear phase trend and is absorbed almost entirely into the fitted slope. At 5 kHz only 0.65 cycles fit and the damage peaks; by 250 kHz some 32 cycles fit, the perturbation averages away, and the contribution vanishes. The same argument applies to the offset d , whose perturbation is at Δf_{BB} itself.

6.6 The accepted window inside the tone

The $80 \mu s$ of T_{FM} is not $80 \mu s$ of usable signal. At its start the transmitter is still settling after the guard interval and the receiver’s automatic gain control is still converging; at its end the transmitter begins to ramp down. Both regions carry amplitude and phase transients that a frequency estimator will faithfully interpret as frequency.

The estimator therefore operates on

$$W = [t_0 + t_{settle}, t_0 + T_{FM} - t_{tail}], \quad T_{obs} = T_{FM} - t_{settle} - t_{tail}, \quad N = \lfloor T_{obs} F_s \rfloor, \quad (23)$$

with t_0 the tone start derived from the timing reference of Section 4.

Table 6: Cost of window trimming. The frequency standard deviation scales as $T_{obs}^{-3/2}$ at fixed signal-to-noise ratio, because both the observation length and the accumulated energy grow with it. Values at $\gamma = 20$ dB in a 1 MHz reference bandwidth.

t_{settle}	t_{tail}	T_{obs}	σ_f	σ_f/f_c at 2.44 GHz
0	0	$80 \mu s$	77.1 Hz	0.0316 ppm
$5 \mu s$	$2 \mu s$	$73 \mu s$	88.4 Hz	0.0362 ppm
$10 \mu s$	$5 \mu s$	$65 \mu s$	105.2 Hz	0.0431 ppm
$20 \mu s$	$5 \mu s$	$55 \mu s$	135.2 Hz	0.0554 ppm

Trimming is not optional, but it is not free either: the penalty from an over-generous margin is real and grows quickly. The correct margins are the *measured* settling behaviour of the actual radio, characterised once and fixed, not a defensive guess.

7 Frequency estimators

The specification defines the signal interval and the quantity to be reported; it does not prescribe an estimator [Core 6.3, Vol. 6, Part A, Sec. 3.5]. What follows is the standard family for a single complex tone in noise, with the properties that matter for this particular 80-microsecond problem.

7.1 Lag- m autocorrelation

The cheapest useful estimator forms the average phase advance over a lag of m samples:

$$\widehat{\Delta f}_{BB} = \frac{F_s}{2\pi m} \arg \left(\sum_{n=m}^{N-1} z[n] z^*[n-m] \right). \quad (24)$$

Taking the argument of the *sum* rather than summing the arguments is what makes this robust: it is a coherent average, so it does not require any per-sample phase unwrapping and it degrades gracefully at low signal-to-noise ratio.

The lag choice is an ambiguity decision, and the tempting value is unsafe. Equation (24) is unambiguous only while $|\Delta f_{BB}| < F_s/(2m)$. Larger m gives a longer phase lever arm and lower variance, so it is tempting to push it. At $F_s = 4$ Msample/s:

m	Unambiguous $ \Delta f_{BB} $	Equivalent at 2.44 GHz
1	± 2.000 MHz	± 819.7 ppm
4	± 500 kHz	± 204.9 ppm
16	± 125 kHz	± 51.2 ppm
64	± 31.2 kHz	± 12.8 ppm

A lag of 64 wraps at ± 12.8 ppm — *inside* the tolerance of a legitimate crystal pair, two of which at ± 12 ppm can legitimately differ by up to 24 ppm. Such an estimator will report a confident, low-variance, wholly wrong answer on a perfectly compliant device. The safe construction is a two-stage one: a small lag to acquire unambiguously, then a larger lag to refine within the interval the first stage has established.

7.2 Unwrapped-phase least squares

Form $\psi[n] = \text{unwrap}(\arg z[n])$ and fit a straight line:

$$\widehat{\Delta f}_{BB} = \frac{F_s}{2\pi} \frac{\sum_n (n - \bar{n})(\psi[n] - \bar{\psi})}{\sum_n (n - \bar{n})^2}. \quad (25)$$

This attains the bound at high signal-to-noise ratio and yields a residual that is a genuinely useful diagnostic: a straight residual indicates a clean tone, a curved one indicates drift or a settling transient inside the window, and a residual with structure at the frame rate indicates interference. Its weakness is the unwrapping, which fails once per-sample phase noise approaches π ; at low signal-to-noise ratio Eq. (24) is the more robust choice.

7.3 Periodogram and Rife–Boorstyn refinement

Take the discrete Fourier transform of the windowed samples, locate the peak bin, and refine locally by interpolation or a Newton step on the periodogram. This is the maximum-likelihood estimator for a single tone in white noise and is the natural reference implementation against which the cheaper two are validated.

Bin spacing is not estimator resolution. The transform’s bin spacing is $1/T_{obs}$, which is 12.5 kHz for an $80\ \mu s$ window — about 5 ppm at 2.44 GHz. That number is not the accuracy of the estimator and should never be quoted as such. It is the spacing of a search grid; the peak’s *location* is estimated from the shape of a smooth function and is resolved to a small fraction of a bin, exactly as sub-sample interpolation resolves a correlation peak in Mode-1. The achievable accuracy is Eq. (26), some 160 times finer than the bin spacing at 20 dB.

7.4 Cramér–Rao bound

For a single complex tone of total energy-to-noise-density $\mathcal{E}/N_0$ observed over T_{obs} ,

$$\sigma_f \geq \frac{1}{\pi T_{obs}} \sqrt{\frac{3}{\mathcal{E}/N_0}}, \quad \frac{\mathcal{E}}{N_0} = \gamma B_n T_{obs}, \quad (26)$$

with γ the signal-to-noise ratio in a reference noise bandwidth B_n . Substituting the second relation into the first shows the $T_{obs}^{-3/2}$ scaling of Table 6. Expressed dimensionlessly, $\sigma_{\text{FFO}} = \sigma_f / f_{nom}$.

Table 7: Frequency and fractional-frequency precision from Eq. (26), with $B_n = 1\text{ MHz}$ and $f_{nom} = 2.44\text{ GHz}$. The last column is the value after combining three Mode-0 steps.

γ	T_{obs}	$\mathcal{E}/N_0$	σ_f	σ_{FFO}	$\sigma_{\text{FFO}}, M=3$
30 dB	$80\ \mu s$	49.0 dB	24.4 Hz	0.0100 ppm	0.0058 ppm
20 dB	$80\ \mu s$	39.0 dB	77.1 Hz	0.0316 ppm	0.0182 ppm
10 dB	$80\ \mu s$	29.0 dB	243.7 Hz	0.0999 ppm	0.0577 ppm
0 dB	$80\ \mu s$	19.0 dB	770.5 Hz	0.3158 ppm	0.1823 ppm
20 dB	$65\ \mu s$	38.1 dB	105.2 Hz	0.0431 ppm	0.0249 ppm
10 dB	$65\ \mu s$	28.1 dB	332.7 Hz	0.1363 ppm	0.0787 ppm

8 Systematic errors, which dominate

Table 7 sets a thermal floor of a few hundredths of a part per million. Almost nothing in a real implementation is that good, and the purpose of this section is to say what actually decides Mode-0 accuracy.

8.1 Front-end offset and image

Both are derived in Section 6 and quantified in Table 5, and both are governed by the same design choice. The offset d of Eq. (21) is a stationary vector at zero baseband frequency, which every estimator in Section 7 sees as a second tone competing with the real one; the image βz_{id}^* is a conjugate copy at $-\Delta f_{BB}$. Placing the tone at an intermediate frequency of 100 kHz or more separates the wanted signal from both and reduces their combined contribution from 0.1021 ppm to 0.0047 ppm.

Where an intermediate-frequency offset is not available, the fallbacks are weaker: estimate d in a quiet interval and subtract it, calibrate the quadrature imbalance per gain state, or high-pass the record — the last accepting that a corner anywhere near the beat frequency distorts the quantity being measured.

8.2 Sign convention

Derived in Section 6.2 and repeated here because it belongs in any error list: an inverted s_{IQ} reverses the correction rather than reducing it. It arises from an odd number of inversions across two independent categories — the $\pm 90^\circ$ quadrature convention and the sense of any real mixing stage — cannot be read off a block diagram, is invisible to every internal quality metric, and is caught only by an end-to-end signed injection test. A channel exchange additionally carries a constant 90° that Mode-0 cannot see but the phase-based modes can.

8.3 Settling and ramp transients

An untrimmed window admits the transmitter’s turn-on and turn-off behaviour. A phase transient of even a fraction of a radian in the first few microseconds, fitted as part of a linear phase ramp over $80\mu s$, produces a frequency error of order (transient phase)/($2\pi T_{obs}$): a 0.2rad disturbance over the first $5\mu s$ can shift the estimate by hundreds of hertz, exceeding the entire thermal budget. This is the mechanism that makes Eq. (23) mandatory rather than tidy.

8.4 Oscillator phase noise

Phase noise integrated over the observation window adds directly to the phase trajectory that Eq. (25) fits. Unlike thermal noise it is correlated in time, so it is not reduced by combining samples within one tone, though it does average across the one-to-three Mode-0 steps of a subevent. Close-in phase noise — offsets below about $1/T_{obs} \approx 12.5\text{ kHz}$ — is the part that matters, because it looks like frequency drift over the window rather than like noise.

8.5 Motion

A time-varying channel does shift the frequency. The Doppler shift is $f_d = v f_{nom}/c$:

Relative velocity	Doppler at 2.44 GHz	Equivalent FFO error
1 m/s (walking)	8.1 Hz	0.0033 ppm
5 m/s (running)	40.7 Hz	0.0167 ppm
30 m/s (vehicular)	244.2 Hz	0.1001 ppm

At pedestrian speeds Doppler is well below the thermal floor and can be ignored. At vehicular speeds it exceeds it by a factor of three and becomes the largest single term. It is also not an error in any meaningful sense — it is a true frequency shift of the received tone — but it is not the oscillator offset that the downstream corrections assume they are receiving, and an implementation operating at speed should either estimate and remove it using the range rate or widen its error budget to admit it.

8.6 Interference

A second signal within the receiver’s measurement bandwidth is the one impairment that can move the estimate arbitrarily far. Gating is the defence: a per-step residual from Eq. (25), an amplitude consistency check, and rejection of any step whose estimate is inconsistent with the others in the same subevent.

Table 8: Mode-0 error sources, ordered by how much attention they typically deserve rather than by magnitude.

Source	Effect	Mitigation	Diagnostic
s_{IQ} sign inversion	Correction reversed; error doubled	End-to-end signed injection test	FFO versus injected offset of each sign
I/Q channel exchange	Same sign inversion, plus a constant 90° invisible to Mode-0	Verify the constant, not only the sign (§6.2)	Reported phase in Mode-2 against a reference
Actuation error not removed	Bias up to the 4 ppm FAE full scale, channel-dependent	Exchange and apply the peer's value	FFO versus channel
FAE table quantisation	Fixed per-channel bias, ≤ 0.0156 ppm; does not average down	None available; budget for it (§5.6)	FFO residual versus channel, constant across subevents
Peer's actuation table stale or mis-indexed	Per-channel bias	Version and validate the table	FFO scatter across channels
Uncalibrated sample rate	First-order scale error on FFO	Use physical F_s , not nominal	Digital injection test
DC offset (d)	Bias toward zero beat; 0.032 ppm at -30 dBc near DC	Intermediate-frequency placement (§6.5)	FFO versus true offset near zero
I/Q imbalance (image)	Conjugate copy at $-\Delta f_{BB}$; 0.083 ppm worst case near DC	Intermediate-frequency placement; per-gain calibration	FFO versus tone position in baseband
Settling transient in window	Hundreds of Hz of bias	Trim per Eq. (23)	FFO versus t_{settle} sweep
Lag ambiguity wrap	Confident wrong answer beyond $\pm F_s/2m$	Two-stage lag	FFO versus large injected offset
Thermal noise	Random, Eq. (26)	Longer window, more steps	Scatter versus predicted CRLB
Phase noise	Correlated within a step	Oscillator budget	Residual shape of the phase fit
Doppler	True shift, 0.1 ppm at 30 m/s	Range-rate compensation or wider budget	FFO versus velocity
Interference	Arbitrary error	Gating on residual and amplitude	Rejected-step rate per channel

9 Combining steps and producing the subevent result

With $M \in \{1, 2, 3\}$ Mode-0 steps in the subevent, each yielding $\widehat{\text{FFO}}[k]$ with an uncertainty estimate σ_k and a validity gate g_k ,

$$\widehat{\text{FFO}}^* = \frac{\sum_k g_k \sigma_k^{-2} \widehat{\text{FFO}}[k]}{\sum_k g_k \sigma_k^{-2}}. \quad (27)$$

Two cautions apply, both arising from the small M .

First, with at most three samples an inverse-variance weighting driven by data-derived σ_k is poorly conditioned; a variance floor, as in the companion reports' aggregation rules, prevents one optimistic step from dominating. Second, and more important, $M \leq 3$ gives almost no outlier rejection. With two valid steps, a disagreement identifies that something is wrong but not which step is at fault. The practical response is to gate hard on per-step quality — residual from Eq. (25), amplitude consistency, ambiguity margin — rather than to rely on cross-step agreement, and to treat a subevent whose steps disagree by more than a few times the predicted σ as uncalibrated rather than as averageable.

Steps within a subevent are on different channels, and that is informative. Actuation error δ is channel-dependent. If the peer's actuation table is stale, mis-indexed, or simply absent, the resulting bias differs between the Mode-0 steps of one subevent. A systematic spread of $\widehat{\text{FFO}}[k]$ that correlates with channel — rather than looking like noise — is the signature, and it is the cheapest available check that the actuation-error exchange is working.

10 What the downstream modes actually require

This is the section that determines whether a Mode-0 implementation is good enough, and it can only be written by looking outward. Each ranging mode converts residual fractional frequency offset into range error through an interval of its own, by the same relation in every case:

$$\delta d = \frac{c \varepsilon_{\text{res}} T_{\text{lever}}}{2}, \quad (28)$$

where T_{lever} is the interval over which that mode integrates the residual — the reflector turnaround B_R for a round-trip-time chain, the reciprocal measurement separation ΔT for a phase chain.

Table 9: Fractional frequency offset accuracy required by each downstream mode, computed from Eq. (28) by holding the clock-scale term to 10% of that chain's total uncertainty as budgeted in the companion reports.

Consumer	T_{lever}	Chain budget	10% target	Required ε_{res}
Mode-1, RTT	224 μs	11.90 cm	11.90 mm	0.354 ppm
Mode-2, phase	190 μs	1.57 cm	1.57 mm	0.055 ppm
Mode-3, timing chain	227 μs	12.82 cm	12.82 mm	0.377 ppm
Mode-3, phase chain	105 μs	1.57 cm	1.57 mm	0.100 ppm

The binding constraint is Mode-2, by a factor of about seven over the round-trip-time modes. The reason is structural rather than incidental: the phase chains have small budgets because they are precise, so a given absolute error consumes far more of their allowance, and Mode-2 additionally has the longest lever arm of the two phase chains because its interlude is not shared with a packet exchange.

Table 10: Mode-0 capability against the binding 0.055 ppm requirement. Thermal values from Table 7; the total combines the thermal term with the FAE quantisation floor of Section 5.6 (0.0090 ppm root-mean-square) in quadrature. ✓ denotes a margin, × a shortfall.

γ	T_{obs}	$M = 1$			$M = 3$		
		thermal	total	vs req.	thermal	total	vs req.
20 dB	80 μs	0.0316	0.0328	✓	0.0182	0.0203	✓
20 dB	65 μs	0.0431	0.0441	✓	0.0249	0.0265	✓
10 dB	80 μs	0.0999	0.1003	×	0.0577	0.0584	×
10 dB	65 μs	0.1363	0.1366	×	0.0787	0.0792	×

Mode-0 is adequate at good signal-to-noise ratio and becomes the limiting factor below about 18 dB. Solving Eq. (26) for the signal-to-noise ratio at which a single trimmed 65 μs window just meets 0.055 ppm gives $\gamma = 18.0$ dB once the FAE quantisation floor is carried in quadrature; with all three Mode-0 steps used, the threshold falls to 13.2 dB. Treating the quantisation instead as the fixed bias it actually is — subtracting its worst case of 0.0156 ppm from the allowance rather than combining it in quadrature — moves the thresholds to 20.8 dB and 16.0 dB respectively, and that is the more defensible pair for a design margin. Below that, phase-based ranging is limited by the quality of its frequency calibration rather than by its own phase estimation — and no amount of effort spent on the phase chain will recover it. A design targeting long range or a lossy environment should therefore treat the Mode-0 step count, and the trimming margins that set T_{obs} , as ranging parameters rather than as housekeeping.

11 Worked numerical example at 2 440 MHz

Take $f_{nom} = 2\,440$ MHz, a true relative oscillator offset $\varepsilon = 12$ ppm, synthesiser a peer actuation error of $\text{FAE} = +0.0625$ ppm — an exact multiple of the 0.03125 ppm table resolution, hence $\delta_R = +152.5$ Hz by Eq. (8) — a local actuation error $\delta_I = -120$ Hz, an LE 2M Mode-0 packet, $T_{GD} = 10\ \mu s$, $T_{RD} = 5\ \mu s$, $T_{IP1} = 40\ \mu s$, $F_s = 4$ Msample/s, and a per-tone signal-to-noise ratio of 20 dB in a 1 MHz reference bandwidth.

11.1 Step timing

From Eq. (1), $T_{step,0} = 2(26) + 2(5) + 40 + 10 + 80 = 192\ \mu s$, in agreement with the companion campaign report.

11.2 The quantities in the beat

$$f_{nom} \varepsilon = 2.44 \times 10^9 \times 12 \times 10^{-6} = 29\,280\ \text{Hz}, \quad (29)$$

$$\delta_R - \delta_I = 152.5 - (-120) = 272.5\ \text{Hz}, \quad (30)$$

$$\Delta f_{BB} = 29\,280 + 272.5 = 29\,552.5\ \text{Hz}. \quad (31)$$

Over the full 80 μs tone this is 2.364 cycles of rotation, or 14.86 rad — ample for a phase fit, and comfortably inside the ± 2 MHz unambiguous range of a lag-1 estimator.

11.3 Reconstruction and FFO

From Eq. (6) with $s_{IQ} = +1$,

$$\hat{f}_{RX} = 2\,440\,000\,000 - 120 + 29\,552.5 = 2\,440\,029\,432.5\ \text{Hz}, \quad (32)$$

and from Eq. (7), with $f_{nom}(1 + 10^{-6}\text{FAE}) = 2\,440\,000\,152.5\text{ Hz}$,

$$\widehat{\text{FFO}}_{RX} = 10^6 \frac{2\,440\,029\,432.5 - 2\,440\,000\,152.5}{2\,440\,000\,152.5} = 12.0000\text{ ppm}, \quad (33)$$

recovering the true offset exactly, as it must in the noise-free case.

11.4 The two failure modes, quantified

Table 11: Effect of the two structural errors on the worked example. The final column converts the error into Mode-2 range through Eq. (28) at $\Delta T = 190\text{ }\mu\text{s}$.

Case	Reported FFO	Error	Mode-2 range error
Correct	12.0000 ppm	—	—
FAE not removed, this example	12.0625 ppm	0.0625 ppm	1.78 mm
FAE not removed, full scale 4 ppm	—	4.0000 ppm	114 mm
FAE quantisation, worst case	—	0.0156 ppm	0.45 mm
s_{IQ} inverted	-12.2234 ppm	24.2234 ppm	690 mm

Every row repays attention. This example’s FAE is deliberately small; the specification permits it to reach 4 ppm, and an implementation that never applies the table at all therefore risks seventy times the entire Mode-2 requirement of Table 9 — from a synthesiser artefact that has nothing to do with either oscillator’s rate. That is the quantitative answer to “why does actuation error have to be exchanged at all”. The quantisation row is the floor that remains after the table *is* applied, and unlike the thermal term it does not average down. Inverting the sign costs 24.2 ppm — twice the true offset — and turns a calibration into an anti-calibration.

11.5 Precision

From Eq. (26) at $\gamma = 20\text{ dB}$ over the full window, $\mathcal{E}/N_0 = 8000\text{ (39.0 dB)}$ and $\sigma_f = 77.1\text{ Hz}$, or 0.0316 ppm . The transform bin spacing over the same window is 12.5 kHz , some 162 times coarser — the standing reminder that bin spacing is a search grid, not an accuracy.

Table 12: Deterministic values for the 2 440 MHz example.

Quantity	Value	Unit
T_{SY} (LE 2M, no sequence) / T_{GD} / T_{RD} / T_{IP1} / T_{FM}	26 / 10 / 5 / 40 / 80	μs
Step duration $T_{step,0}$	192	μs
True relative oscillator offset	12.0000	ppm
$f_{nom}\varepsilon$	29 280	Hz
Peer actuation error FAE	+0.0625	ppm
Actuation errors δ_R / δ_I	+152.5 / -120	Hz
Measured beat Δf_{BB}	29 552.5	Hz
Rotation over the 80 μs tone	14.86	rad
Reconstructed f_{RX}	2 440 029 432.5	Hz
Recovered FFO	12.0000	ppm
Error if FAE not removed (this example)	0.0625	ppm
Error if FAE not removed (full scale)	4.0000	ppm
FAE quantisation, worst case	0.015625	ppm
FAE quantisation, rms	0.0090	ppm
Error if s_{IQ} inverted	24.2234	ppm
$\mathcal{E}/N_0$ at $\gamma = 20$ dB, 80 μs	39.0	dB
σ_f	77.05	Hz
σ_{FFO} , $M = 1$	0.0316	ppm
σ_{FFO} , $M = 3$	0.0182	ppm
Transform bin spacing $1/T_{obs}$	12.5	kHz
Bin spacing in ppm	5.123	ppm
Binding downstream requirement (Mode-2)	0.055	ppm
SNR threshold, $M = 1$, $T_{obs} = 65 \mu s$ (quadrature)	18.0	dB
SNR threshold, $M = 3$, $T_{obs} = 65 \mu s$ (quadrature)	13.2	dB
SNR threshold, $M = 1$ (quantisation as bias)	20.8	dB
SNR threshold, $M = 3$ (quantisation as bias)	16.0	dB

12 Verification and conformance-oriented test plan

12.1 Layered verification

1. **Signed injection, before anything else.** Synthesise a tone at a known offset of *each* sign and verify that the reported FFO matches in magnitude *and* sign. This test exists to catch s_{IQ} inversion, and it must be run first: every accuracy characterisation performed before it is meaningless, because an inverted implementation passes all of them.
2. **Constant-rotation check.** A signed injection test alone cannot separate a quadrature polarity inversion from an exchange of the in-phase and quadrature channels, because Mode-0 sees both as $-\Delta f_{BB}$ and absorbs the exchange's constant 90° into θ_0 (Section 6.2). Verify the constant separately, against a reference of known absolute phase or through the phase correction terms of a Mode-2 step, before concluding that a corrected sign means a correct front end.
3. **Pure DSP vectors.** Noise-free tones swept across the expected offset range, with controlled DC offset, I/Q imbalance and clipping. Verify the estimator's bias against injected offset, and characterise its dependence on DC offset magnitude, particularly near zero beat.
4. **Intermediate-frequency placement sweep.** With a fixed injected offset, sweep the local oscillator so that the tone moves across baseband from a few kilohertz to several hundred, and record the estimation error against Table 5. This is the measurement that justifies the chosen f_{IF} , and it exposes both the offset and the image in one sweep. Confirm the non-monotonic image behaviour predicted in Section 6.5.
5. **Ambiguity sweep.** Sweep the injected offset well beyond $\pm F_s/(2m)$ for the implemented lag and confirm the wrap occurs exactly where predicted, and that the two-stage acquisition of Section 7 prevents it. Include ± 24 ppm, the legitimate worst case for a compliant crystal pair.
6. **Thermal consistency.** At $\gamma = 0, 10, 20$ and 30 dB with 10^3 trials each, the sample standard deviation must lie within a documented margin of Table 7: 770.5, 243.7, 77.1 and 24.4 Hz over the full window.
7. **Window characterisation.** Sweep t_{settle} and t_{tail} and record the estimate versus margin. The correct operating point is where the bias has flattened; the curve also documents the transient behaviour of the actual radio, which is the only defensible basis for the chosen margins.
8. **Sample-rate calibration.** Inject digitally at a known offset with a deliberately mis-set nominal F_s and confirm the resulting first-order scale error, then confirm that using the calibrated physical rate removes it.
9. **Actuation-error exchange.** With a peer whose actuation error is known and non-zero, verify that the reported FFO is independent of it. Sweep across channels: a residual that correlates with channel indicates the table is absent, stale or mis-indexed. Then deliberately withhold or corrupt the table and confirm the resulting bias matches $(\delta_R - \delta_I)/f_{nom}$.
10. **End-to-end against a reference.** Offset one device's reference by a calibrated ppm value and verify the reported FFO across the value's range and both signs.
11. **Downstream closure.** Inject a known offset and verify that Mode-1's rescaling, Mode-2's drift correction and Mode-3's both behave as their companion reports predict. This

is the test that establishes the whole chain works, and it is the only one that would catch a self-consistent but wrongly-scaled FFO.

12. **Doppler.** Sweep relative velocity and confirm the reported offset shifts by vf_{nom}/c — 8.1 Hz per m/s at 2.44 GHz — and that the implementation’s handling of it, whether compensation or budget allowance, behaves as documented.
13. **Interference and gating.** Inject an in-band interferer at controlled levels and verify the per-step gate rejects the affected steps rather than averaging them in.
14. **Step-count regression.** Verify behaviour with one, two and three Mode-0 steps, including the case where some are gated out, and that the reported FFO* reflects only the accepted steps.

12.2 Minimum acceptance records

Retain: RF configuration and channel list; PHY; T_{IP1} and step count; calibrated F_s ; the s_{IQ} convention as verified, not as assumed; t_{settle} and t_{tail} with the sweep that justifies them; the peer’s actuation-error table with its version and the local table alongside it; per-step raw beat estimate, residual, amplitude, gate decision and σ_k ; the combined FFO* with accepted step count and scatter; and the downstream-closure result.

13 Normative behaviour versus implementation choice

Table 13: Boundary between specified behaviour and implementation design.

Bluetooth-defined behaviour	Implementation-specific design
Mode-0 step structure and ordering [Core 6.3, Vol. 6, Part H, Sec. 4.3.1]	Receive window placement; capture and buffering
$T_{FM} = 80 \mu s$; T_{GD} , T_{RD} fixed; T_{SY} from packet construction; T_{IP1} value set	The accepted window inside T_{FM} : t_{settle} , t_{tail} , T_{obs}
That the tone is unmodulated at the channel’s commanded frequency [Core 6.3, Vol. 6, Part A, Sec. 3.5]	Intermediate-frequency placement, sample rate, numeric format
Definition of $f_{RX}[k]$ as the average received tone frequency	The frequency estimator, its lag or transform structure, and its refinement
Requirement that the receiver compensate its own actuation error	How the local actuation error is determined and stored
Exchange of the peer’s actuation error, and its use in the FFO relation	Table storage, versioning, staleness policy, interpolation across channels
Number of Mode-0 steps per subevent (one to three)	How many are configured, and the gating and combining rule
Per-step Mode-0 reporting [Core 6.3, Vol. 4, Part E]	Quality metrics, thresholds, uncertainty estimates
That the result calibrates later steps in the subevent	Filtering of FFO* across subevents; drift tracking

Four conclusions that prevent common design errors. (1) The measured beat contains a synthesiser artefact as well as an oscillator offset, and the two are separated by exchanged knowledge rather than by estimation — omitting the peer’s actuation error costs 0.123 ppm in the worked example, more than twice what Mode-2 can tolerate. (2) The in-phase/quadrature sign convention must be verified by signed injection before any accuracy work; an inversion is invisible to every internal metric and doubles the error. (3) Transform bin spacing is a search grid, not an accuracy; at 20 dB the estimator resolves the peak some 160 times finer than the bins. (4) A lag chosen for low variance can wrap inside the tolerance of a compliant crystal pair — lag-64 at 4 Msample/s wraps at ± 12.8 ppm — so acquisition and refinement must be separate stages.

14 Conclusions

Mode-0 is the smallest of the Channel Sounding step modes and the one on which every other depends. Its physical content is a single comparison: an unmodulated tone from one device, measured against the other device’s own reference, over eighty microseconds. Everything that makes it seem complicated comes from one fact — that the beat frequency contains two contributions of entirely different origin, only one of which is wanted — and from the consequence that the unwanted one can be removed only by the peer telling you its value.

The estimation problem itself is the textbook single-tone problem and is comfortably solved. The Cramér–Rao bound over the full tone at 20 dB is 0.032 ppm, and the binding downstream requirement, set by Mode-2 phase-based ranging, is 0.055 ppm. Mode-0 therefore has margin at good signal-to-noise ratio, and loses it below about 18 dB with a single step or 13 dB with three — at which point the precision of phase-based ranging is limited by its frequency calibration and not by its own phase estimator. That crossover, rather than the bound itself, is the number an implementer should carry.

What decides Mode-0 accuracy in practice is not the estimator. It is whether the sign convention was verified rather than assumed, whether the peer’s actuation error is actually being applied, whether the observation window excludes the transmitter’s transients, and whether the sample rate used to convert cycles per sample into hertz is the physical one. Each of those is worth more than any refinement of the frequency estimator, and three of the four are invisible to every quality metric the estimator can produce. A Mode-0 implementation should be validated from the outside — by injecting known offsets of both signs and confirming that the ranging modes downstream behave as their own budgets predict — because that is the only test the structural failures cannot pass.

A Reported quantities at the host interface

A Mode-0 step reports a small field set, with the initiator reporting the frequency result and the reflector reporting only reception quality, reflecting the asymmetry of the step [Core 6.3, Vol. 4, Part E, LE CS Subevent Result].

Two implementation notes. First, the frequency result is reported as a dimensionless offset, not as a frequency in hertz, so the scaling and resolution must be confirmed rather than inferred — an off-by-a-decade unit interpretation is a plausible and catastrophic error that no internal check would catch. Second, the reserved “unavailable” value must be tested for explicitly; interpreted as a numeric offset it does not merely lose one step, it mis-calibrates every ranging step in the subevent that follows.

Table 14: Per-step Mode-0 reported fields. Field names, widths, units and resolutions must be read from the specification revision in force; they could not be confirmed from public sources during preparation and are described here by role.

Field	Reported by	Role
Measured frequency offset	Initiator only	The Mode-0 result; feeds every later step in the subevent
Packet quality	Both	Gating input for the CS_SYNC reception
Packet RSSI	Both	Signal-to-noise proxy; gating input
Packet antenna	Both	Calibration selection
Unavailable indicator	Both	Step must be gated out

B Numerical test vectors

- D1. Signed reconstruction.** With $f_{nom} = 2.44$ GHz, $\delta_I = -120$ Hz and a measured beat of $+29\,552.5$ Hz, verify $\hat{f}_{RX} = 2\,440\,029\,432.5$ Hz. Repeat with the beat negated and confirm the reported FFO changes sign and not only magnitude.
- D2. Actuation-error removal.** With $F_{AE} = +0.0625$ ppm, verify the reported FFO is 12.0000 ppm. Withhold the FAE table and verify the error is exactly 0.0625 ppm. Repeat at the FAE extremes of -4 and $+3.96875$ ppm and confirm the errors are 4.0000 and 3.96875 ppm respectively.
- D3. FAE encoding.** Verify the table is interpreted as 8-bit signed values of 0.03125 ppm each, that -128 maps to -4 ppm and $+127$ to $+3.96875$ ppm, and that the entries are indexed over the allowed CS channels only, shifted down to fill the void indices [Core 6.3, Vol. 6, Part B, Sec. 2.4.2.52]. Inject a table offset by one channel index and confirm the resulting FFO error correlates with channel.
- D4. Sign inversion.** Force $s_{IQ} = -1$ and verify the reported FFO is -12.2234 ppm, an error of 24.2234 ppm. This vector documents the failure the implementation must not have.
- D5. Rotation and ambiguity.** Confirm the $29\,552.5$ Hz beat produces 14.86 rad over $80\,\mu s$, and that lag-1, lag-4 and lag-16 estimators all agree while lag-64 wraps once the injected offset exceeds 12.8 ppm.
- D6. Thermal.** At $\gamma = 20$ dB over $80\,\mu s$, 10^3 trials, verify $\sigma_f = 77.1$ Hz and $\sigma_{FFO} = 0.0316$ ppm; at $65\,\mu s$, 105.2 Hz and 0.0431 ppm.
- D7. Window.** Sweep t_{settle} from 0 to $20\,\mu s$ with a synthetic 0.2 rad turn-on transient in the first $5\,\mu s$ and verify the bias falls as the transient is excluded, and that σ_f rises per Table 6.
- D8. DC offset.** With a true beat of 500 Hz, sweep the DC offset from 0 to the tone amplitude and record the bias toward zero. Repeat with the tone at a non-zero intermediate frequency and confirm the bias is removed.
- D9. Step combining.** With three steps of unequal quality, verify Eq. (27) weights correctly, that a gated step is excluded, and that a channel-correlated spread is flagged rather than averaged.
- D10. Downstream closure.** Inject $\varepsilon = +12$ ppm and verify that Mode-1’s rescaling removes 40.3 cm, that Mode-2’s correction removes the drift its report predicts, and that both residuals fall below their budgeted values. Repeat at -12 ppm.

D11. Step duration. Verify $T_{step,0} = 192\,\mu s$ for the LE 2M configuration of Section 11, and $228\,\mu s$ on LE 1M.

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